

3. For the inductive element of Fig. 15.79:

a. Write the current in phasor form.

$$I_L = 7.071 \text{ mA} \angle 50^\circ$$

b. Calculate the voltage across the inductor in phasor form.

$$V_L = 14.142 \angle 40^\circ \text{ V}$$

c. Sketch the phasor diagram of the voltage and current.

d. Write the voltage in the sinusoidal format.

$$V_L(t) = 20 \sin(250t + 130^\circ)$$

e. Sketch the waveform of the voltage and current.

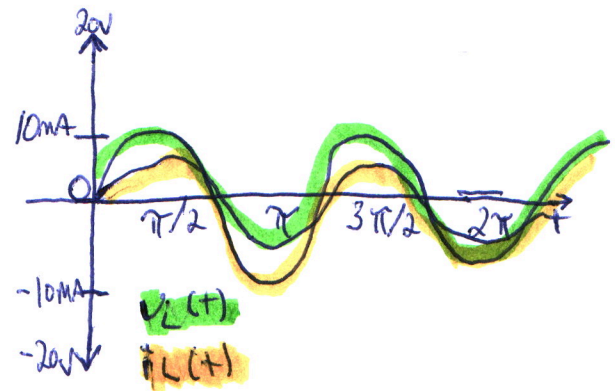
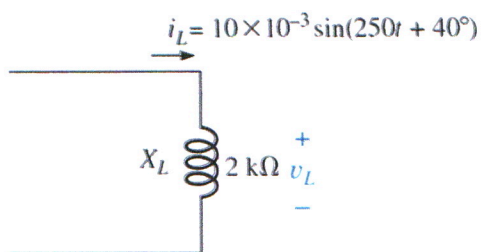


Fig. 15.79



5. For the inductive element of Fig. 15.81:

a. Write the voltage and current in phasor form.

Current: $I_L = 4.243 \text{ mA} \angle -70^\circ$ Voltage: $V_L = 11.314 \angle 20^\circ \text{ V}$

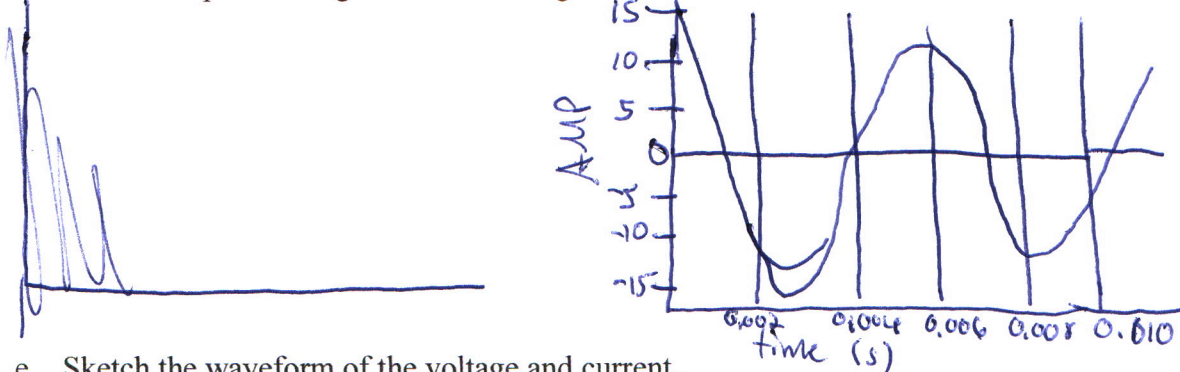
b. Calculate the impedance of the inductor.

$j2666.7 \Omega$

c. Find the inductance of the coil.

$L = 2.22 \text{ H}$

d. Sketch the phasor diagram of the voltage and current.



e. Sketch the waveform of the voltage and current.

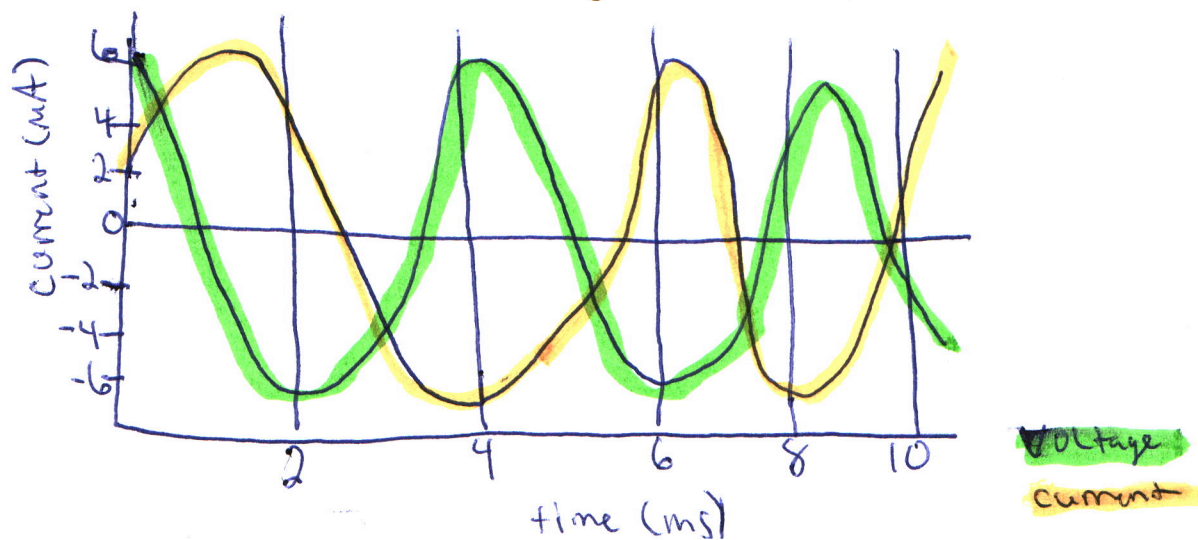
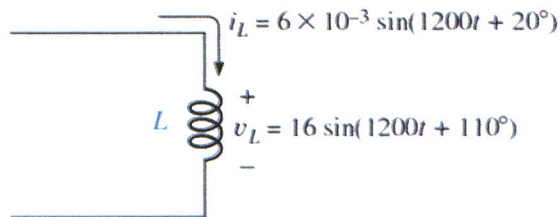


Fig. 15.81



7. For the capacitive element of Fig. 15.83:

- a. Calculate the reactance of the capacitor

$$X_C = 5k\Omega$$

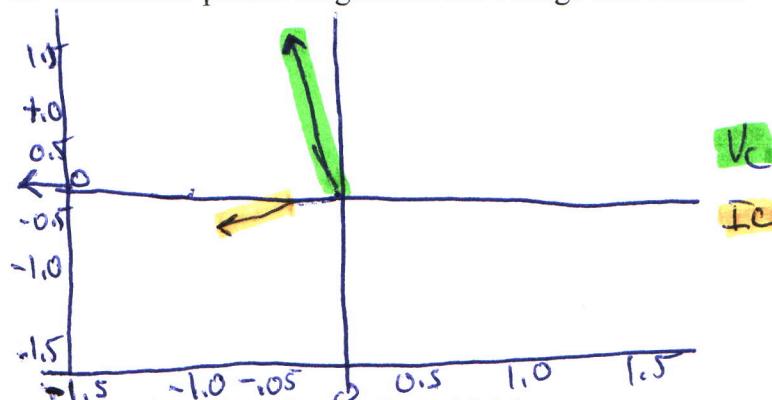
- b. Write the current in phasor form.

$$I_C = 3.536 \mu A \angle -170^\circ$$

- c. Calculate the voltage across the capacitor in phasor form.

$$V_C = 17.68 \text{ mV} \angle 100^\circ$$

- d. Sketch the phasor diagram of the voltage and current.



- e. Write the voltage in the sinusoidal format.

$$V_C(t) = 25 \text{ mV} \sin(20,000t - 170^\circ)$$

- f. Sketch the waveform of the voltage and current.

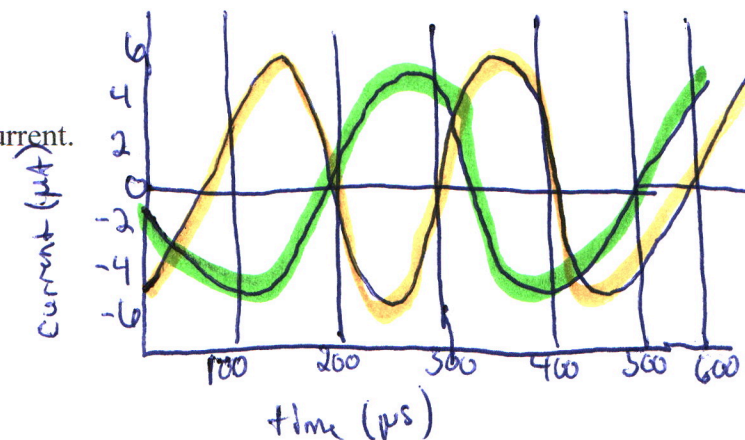
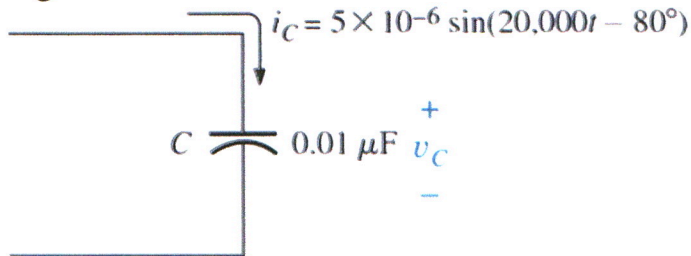


Fig.15.83



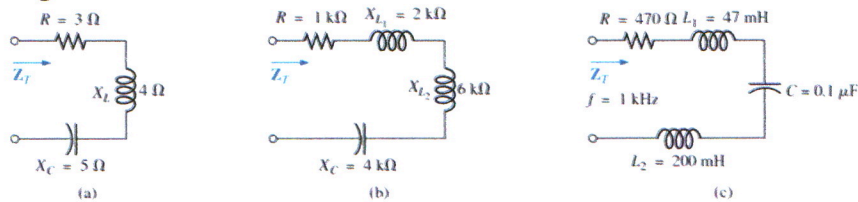
13. Calculate the total impedance of the circuits in Fig. 15.86. Express your answer in rectangular and polar forms, and draw the impedance diagram.

a. $Z_T = 3 - j1 \Omega = 3.162 \angle -18.43^\circ \Omega$

b. $Z_T = 1000 - j4000 \Omega = 4.123 \text{ k}\Omega \angle 75.96^\circ$

c. $Z_T = 470 - j39.60 \Omega = 471.67 \angle -4.82^\circ \Omega$

Fig. 15.86



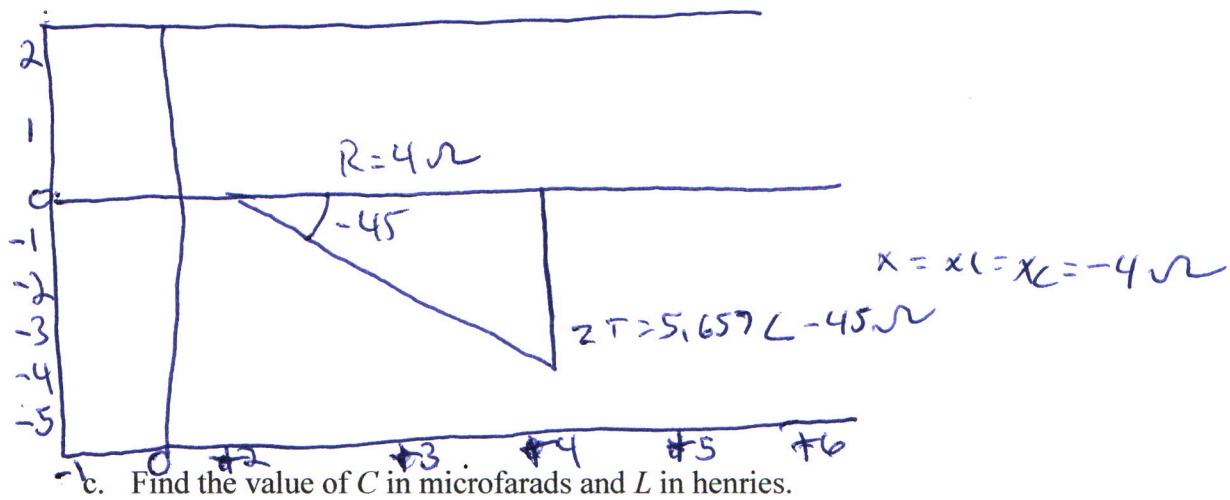
17. For the circuit in Fig. 15.90:

a. Find the total impedance Z_T in polar form.

$Z_T = 5.657 \angle 45^\circ \Omega$

b. Draw the impedance diagram.

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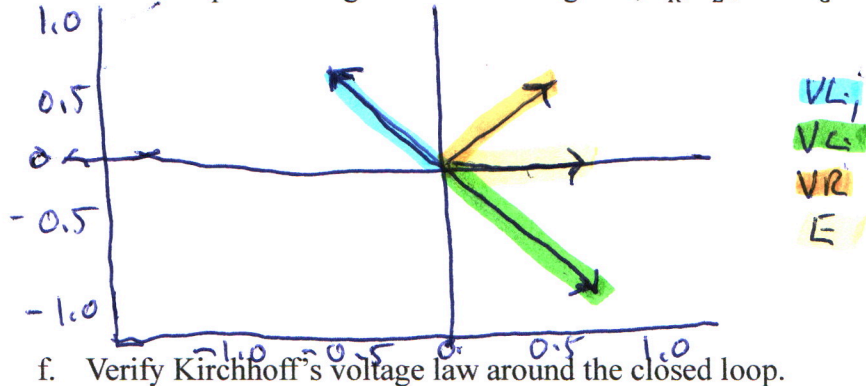


$$L = 15.9 \text{ mH} \quad C = 265.3 \text{ } \mu\text{F}$$

d. Find the current I and the voltages V_R , V_L , and V_C in phasor form.

$$I = 8.839 \angle -45^\circ \text{ V} \quad V_R = 35.36 \angle -45^\circ \text{ V} \quad V_L = 53.03 \angle 45^\circ \text{ V} \quad V_C = 88.39 \angle -135^\circ \text{ V}$$

e. Draw the phasor diagram of the voltages E , V_R , V_L , and V_C and the current I .



$$V_R + V_L + V_C = E$$

g. Find the average power delivered to the circuit.

$$P = 312.5 \text{ W}$$

h. Find the power factor of the circuit, and indicate whether it is leading or lagging.

$$\text{Pf} = 0.707 \text{ leading}$$

i. Find the sinusoidal expressions for the voltages and current.

$$i(t) = 12.5 \sin(377t + 45^\circ) \text{ V}, \quad V_R(t) = 50 \sin(377t + 45^\circ) \text{ V}, \quad V_L(t) = 75 \sin(377t - 45^\circ) \text{ V}$$

$$V_C(t) = 125 \sin(377t - 45^\circ) \text{ V}$$

j. Plot the waveform for the voltages and current on the same set of axes.

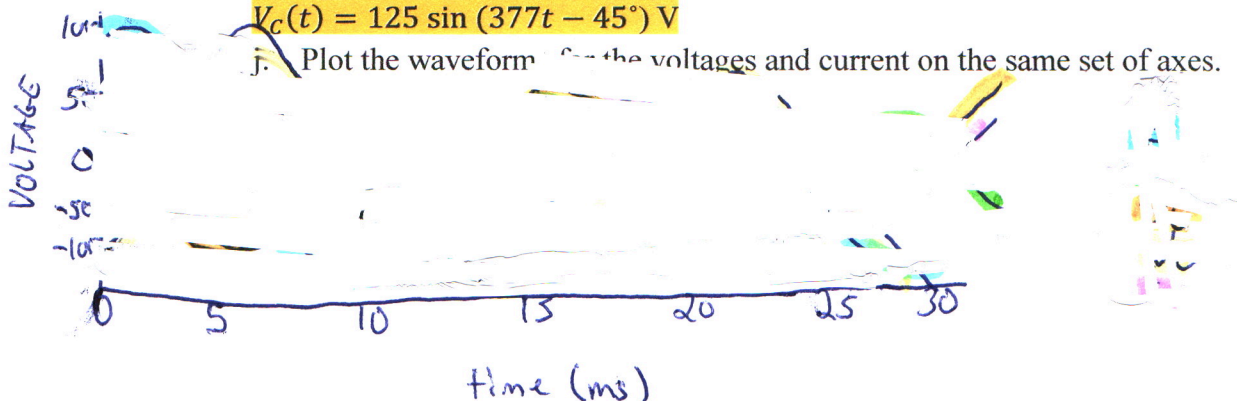
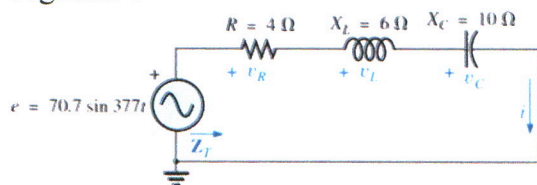




Fig. 15.90



21. For the circuit of Fig. 15.94:

a. Find the current I_{L_1}

$$I_{L_1} = 5 \text{ mA} \angle 30^\circ$$

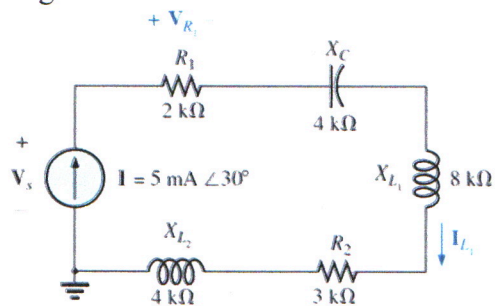
b. Find the voltage V_8

$$V_8 = 47.17 \angle 88^\circ \text{ V}$$

c. Calculate the voltage V_{R_1}

$$V_{R_1} = 8.660 + j5.000 \text{ V}$$

Fig. 15.86

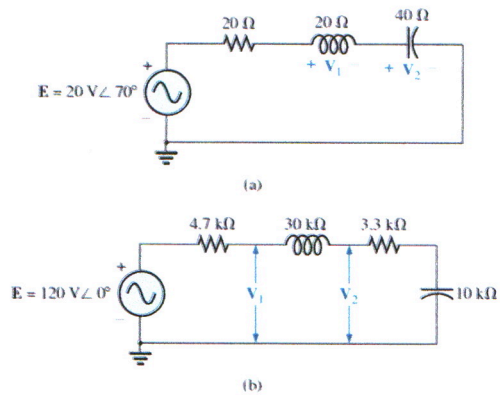


28. Calculate the voltages V_1 and V_2 for the circuits in Fig. 15.100 in phasor form using the voltage divider rule.

a. $V_1 = 14.14 \angle -155^\circ \text{ V}$, $V_2 = 28.28 \angle 25^\circ \text{ V}$

b. $V_1 = 112.92 \angle 12.43^\circ \text{ V}$, $V_2 = 58.66 \angle -139.94^\circ \text{ V}$

Fig. 15.100



3-a) $I_m = 10 \text{ mA}$

$$I = \frac{10}{\sqrt{2}} \text{ mA} = 7.071 \text{ mA}$$

$$40^\circ - 90^\circ = -50^\circ$$

$$I_L = 7.071 \text{ mA} \angle -50^\circ$$

b) $V_L = j\omega L I = 2000 (0.007071) = 14.142 \text{ V} \quad -50^\circ + 90^\circ = 40^\circ$

$$V_L = 14.142 \angle 40^\circ \text{ V}$$

d) $V_m = \sqrt{2} (14.142) = 20 \text{ V} \quad 40^\circ + 90^\circ = 130^\circ$

$$V_L(t) = 20 \sin(250t + 130^\circ) \text{ V}$$

5. a) $I = \frac{6 \cdot 10^{-3}}{\sqrt{2}} \angle (20^\circ - 90^\circ) \quad I_L = 4.243 \text{ mA} \angle -70^\circ$

$$V = \frac{16}{\sqrt{2}} \angle (110^\circ - 90^\circ) = V_L = 11.314 \angle 20^\circ \text{ V}$$

b) $Z_L = \frac{V_L}{I_L} \rightarrow \frac{11.314}{0.004243} = 2666.7 \Omega \quad 20^\circ - (-70^\circ) = 90^\circ$

$$Z_L = 2666.7 \angle 90^\circ \Omega = j2666.7 \Omega$$

c) $\frac{X_L}{\omega} = \frac{2666.7}{1200} = 2.222 \text{ H} \rightarrow L = 2.22 \text{ H}$

7. a) $X_C = \frac{1}{(2000)(1 \cdot 10^{-8})} = 50000 \Omega \rightarrow 50 \text{ k}\Omega$

b) $I_C = \frac{5 \cdot 10^{-6}}{\sqrt{2}} \angle (-80^\circ - 90^\circ) \quad I_C = 3.536 \text{ }\mu\text{A} \angle -170^\circ$

c) $5000 (3.536 \cdot 10^{-6}) = 0.01768 \text{ V} \quad -170^\circ - 90^\circ = -260^\circ = 100^\circ$

$$V_C = 17.68 \text{ mV} \angle 100^\circ$$

e) $V_m = \sqrt{2} (17.68 \text{ mV}) = 25 \text{ mV} \quad 100^\circ + 90^\circ = 190^\circ \quad V_C = 25 \text{ mV} \sin(2000t - 170^\circ)$

13. a) $Z_T = \sqrt{3^2 + (-1)^2} = \sqrt{10} = 3.162 \quad \theta = \tan^{-1}(\frac{-1}{3}) = -18.43^\circ$

$$Z_T = 3.162 \angle -18.4^\circ$$

b) $Z_T = \sqrt{1000^2 + 4000^2} = 4123.1 \Omega \quad \theta = \tan^{-1}(\frac{4000}{1000}) = 75.96^\circ$

$$Z_T = 1000 + j4000 \Omega = 4.123 \text{ k}\Omega \angle 75.9^\circ$$

c) $Z_T = \sqrt{470^2 + (-39.60)^2} = 471.67 \Omega \quad \theta = \tan^{-1}(\frac{-39.60}{470}) = -4.82^\circ$

$$Z_T = 470 - j39.60 \Omega = 471.67 \angle -4.82^\circ$$

17. a) $Z_T = 4 + j6 - j10 = 4 - j4$

$$= \sqrt{4^2 + (-4)^2} = 5.657$$

$$\theta = \tan^{-1}(\frac{-4}{4}) = -45^\circ$$

$$Z_T = 5.657 \angle -45^\circ \Omega$$

$$17. c) L = \frac{6}{377} = 0.0159 \text{ H} \quad \boxed{L = 15.9 \text{ mH}}$$

$$X_C = \frac{1}{\omega C} = \frac{1}{377 \cdot 10} = 2.6525 \cdot 10^{-4} \text{ F} \rightarrow \boxed{265.3 \mu\text{F}}$$

$$d) \text{ current} \rightarrow I = \frac{50 \angle -90^\circ}{5.657 \angle -45^\circ} \quad \boxed{I = 8.839 \angle -45^\circ \text{ A}}$$

$$\text{RESISTOR VOLTAGE} = V_R = IR = 8.839 \angle -45^\circ (4) = \boxed{35.36 \angle -45^\circ \text{ V}}$$

$$\text{CAP INDUCTOR VOLTAGE} = V_L = I(jX_L) = 8.839 \angle -45^\circ (10 \angle 90^\circ) \quad \boxed{V_L = 88.39 \angle -135^\circ \text{ V}}$$

$$\text{IND CAPACITOR VOLTAGE} = V_C = I(-jX_C) = 8.839 \angle -45^\circ (6 \angle 90^\circ) \quad \boxed{V_C = 53.03 \angle 45^\circ \text{ V}}$$

$$f.) (25 - j25) + (37.5 + j37.5) + (-62.5 - j62.5)$$

$$0 - j50 = 50 \angle -90^\circ = E \quad \boxed{V_R + V_L + V_C = E}$$

$$g) P = I^2 R = (8.839)^2 (4) \quad \boxed{P = 312.5 \text{ W}}$$

$$h) \text{ pf} = \cos \theta = \cos(-45^\circ) = 0.707 \rightarrow \text{PF} = 0.707 \text{ lead current leads voltage}$$

$$i) I_m = \sqrt{2} (8.839) = 12.5 \text{ A} \quad -45^\circ + 90^\circ = 45^\circ \quad \boxed{i(t) = 12.5 \sin(377t + 45^\circ) \text{ A}}$$

$$V_{Rm} = \sqrt{2} (35.36) = 50 \text{ V} \quad -45^\circ + 90^\circ = 45^\circ \quad \boxed{V_R(t) = 50 \sin(377t + 45^\circ) \text{ V}}$$

$$V_{Cm} = \sqrt{2} (88.39) = 125 \text{ V} \quad -135^\circ + 90^\circ = -45^\circ \quad \boxed{V_C(t) = 125 \sin(377t - 45^\circ) \text{ V}}$$

$$21. Z_T = (2k + 3k) + j8k + j4k - j4k$$

$$Z_T = 5k + j8k \quad \boxed{I_{L1} = 5 \text{ mA} \angle 30^\circ}$$

$$b) Z_T = (R_1 + R_2) + jX_{L1} + jX_{L2} - jX_{C1} \quad V_S = I_{Z_T} \quad V_S = (5 \text{ mA} \angle 30^\circ)(9.434k \angle 58.0^\circ)$$

$$= \sqrt{(5k)^2 + (8k)^2} = 9.434k \angle \theta$$

$$\theta = \tan^{-1}\left(\frac{8k}{5k}\right) = 58^\circ$$

$$V_S = 47.17 \angle 88.0^\circ \text{ V}$$

$$\boxed{V_S = 47.17 \angle 88.0^\circ \text{ V}}$$

$$28. c) Z_T = 20 + j20 - j40 = 20 - j20$$

$$V_1 = 20 \angle 70^\circ \left(\frac{j20}{20 - j20} \right) = 14.14 \angle -15^\circ \text{ V}$$

$$V_2 = 20 \angle 70^\circ \left(\frac{-j40}{20 - j20} \right) = 28.28 \angle 25^\circ \text{ V}$$

$$b) V_1 = E \left(\frac{j3000 + j3300 - j10000}{8000 + j20000} \right) \quad \boxed{V_1 = 112.52 \angle 12.43^\circ \text{ V}}$$

$$V_2 = E \left(\frac{3300 - j10000}{8000 + j20000} \right)$$

$$\boxed{V_2 = 58.66 \angle -139.94^\circ \text{ V}}$$